

EXPT NO:3	CLEANSE DATA BY HANDLING MISSING VALUES, OUTLIERS, AND INCONSISTENCIES USING TABLEAU AND POWERBI
DATE: 10.07.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

- 1. What are the potential impacts of missing values in a dataset, and what are some common strategies to address them in Tableau and Power BI?**

Missing values can reduce data accuracy, produce incorrect analysis, and lead to misleading results. In Tableau and Power BI, they can be handled by replacing missing numeric values with the mean or median, replacing missing text values with "Unknown" or "Not Applicable," or using calculated fields such as IFNULL() in Tableau and IF(ISBLANK()) in Power BI.

- 2. What is the importance of identifying and addressing outliers in a dataset? How can outliers affect data analysis and model performance?**

Outliers are unusual data values that can distort analysis and reduce the accuracy of reports and predictive models. They can be identified using scatter plots or box plots and managed by filtering them, applying value caps, or flagging them for further review.

- 3. What types of data inconsistencies might you encounter during data cleansing, and how can Tableau and Power BI help in resolving them?**

Common inconsistencies include spelling variations, inconsistent text values, incorrect date formats, and different currency or number formats. Tableau resolves these issues using **Group**, **Alias**, and calculated fields, while Power BI uses **Replace Values**, **Power Query**, and data transformation tools to standardize the data.

- 4. How do you identify patterns in data using visualizations to spot missing values, outliers, and inconsistencies?**

Visualizations such as histograms, bar charts, scatter plots, and box plots help identify missing values, unusual data points, and inconsistent patterns. Comparing charts before and after data cleaning helps verify that the dataset has been corrected successfully.

- 5. What is the role of calculated fields in data cleansing, and how can they be used to handle missing values and outliers in Tableau/Power BI?**

Calculated fields automate data cleaning by replacing missing values, creating standardized values, and flagging outliers. In Tableau, functions such as IFNULL() can replace missing values, while in Power BI, IF(ISBLANK()) can be used to perform similar operations. Calculated fields also help identify records that require further review.

IN-LAB EXERCISE:

OBJECTIVE:

To clean a given dataset (e.g., Superstore Sample in Tableau and Financials/Sales Data in Power BI) by identifying and resolving missing values, outliers, and inconsistencies using built-in tools, ensuring the dataset is accurate and ready for analysis.

PROCEDURE:

1. Open the Tool and Connect Dataset

- Open Tableau Desktop , Power BI Desktop.
- Load the built-in dataset (Sample - Superstore in Tableau / Financials in Power BI).
- Preview the dataset to understand its structure.

2. Initial Inspection ○ Explore fields such as Sales, Profit, Region, or Product Category.

- Check for nulls, inconsistent text values (e.g., "South Region" vs. "South"), or date formatting issues.

3. Handling Missing Values ○ Use Data Interpreter (Tableau) / Power Query (Power

BI) to highlight nulls. ○ Replace missing numerical values with mean/median, and text with "Unknown".

- Use calculated fields: IFNULL([Field], Default) in Tableau or IF(ISBLANK([Field]), "Unknown", [Field]) in Power BI.

4. Detecting and Managing Outliers ○ Create scatter or box plots for Sales and

Profit to identify extreme values.

- Choose strategies:
 - Filter them temporarily.
 - Apply caps using calculated columns.
 - Flag them with labels for review.

5. Resolving Inconsistencies

- Use Group/Alias (Tableau) / Replace Values (Power BI) to standardize text.
- Normalize data formats, such as date fields or currency units.

Validation with Visuals

- Build dashboards with histograms, bar charts, and box plots to visually confirm improvements.
- Compare before/after cleaning stages for accuracy.

2. Save the Cleaned Dataset

- Save as .twbx (Tableau) / .pbix (Power BI).
- Optionally export cleaned data to .csv.

3. Summary Dashboard

- Create a visual summary of the cleaning process, highlighting:
 - Fields corrected.
 - Nulls/outliers removed or adjusted.
 - Clean dataset metrics (record count, completeness %).

Experiment Dataset:

1. Tableau: *Sample - Superstore*
2. Power BI: *Financials* or *Sales Data*

How to Access:

Tableau:

- **Navigate to:** *Start Page → Sample Workbooks*
- Or: *Help → Sample Data → Open Sample Data Folder*

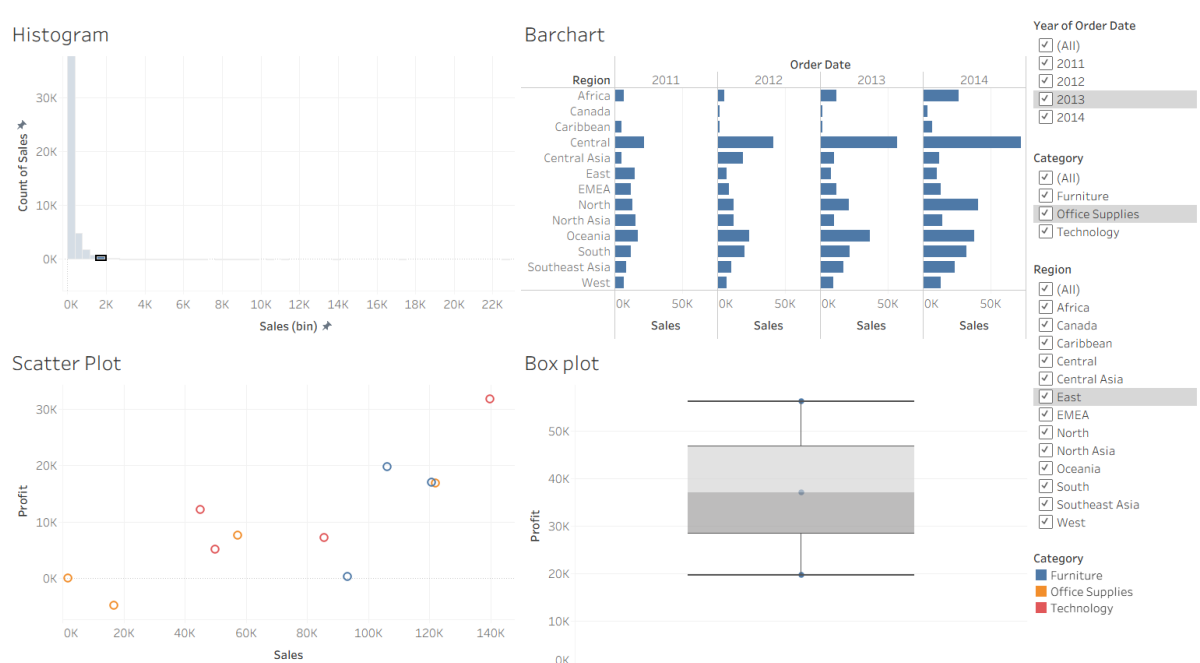
Power BI:

- **Navigate to:** *Home → Get Data → Excel → Choose sample datasets (e.g., Financial Sample.xlsx)*
- Microsoft also hosts Power BI sample datasets online.

Fields Used:

1. Sales
2. Profit
3. Region
4. Category
5. Order Date / Transaction Date

OUTPUT :



POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. What method did you use to replace missing values in numeric fields?

Missing numeric values were replaced using appropriate default values such as the mean or median, while missing text values were replaced with "Unknown" or "Not Applicable" using calculated fields.

2. How were outliers handled in your dataset and what visual confirmed their correction?

Outliers were identified using scatter plots and box plots. They were flagged using a calculated field and filtered for analysis. The box plot confirmed the correction by showing fewer extreme values after cleaning.

3. What inconsistencies were detected in categorical values and how were they standardized?

Inconsistencies such as spelling variations and inconsistent category names were identified. They were standardized using Group, Alias, or Replace Values to ensure consistent categorical values throughout the dataset.

4. Describe how calculated fields helped in resolving data issues.

Calculated fields were used to replace missing values, create standardized values, and flag outliers. They automated the data-cleaning process and improved the consistency and accuracy of the dataset.

5. How did your dashboard demonstrate the effectiveness of the data cleaning process?

The dashboard combined visualizations such as histograms, bar charts, scatter plots, and box plots to verify the cleaned data. Interactive filters and charts showed reduced missing values, standardized categories, and properly identified outliers, confirming that the dataset was ready for analysis.

RESULT:

The dataset was successfully cleaned and prepared for accurate analysis using Tableau and Power BI. Issues such as nulls, inconsistencies, and outliers were identified and resolved, improving data quality significantly.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		